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Portfolio Optimization: Asset Universe Selection Based on Investor Profiles

Bachelor Thesis

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This is to certify that:

(i) the thesis comprises only my original work toward the Bachelor Degree

(ii) due acknowledgement has been made in the text to all other material used

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8 June, 2026

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Abstract

Portfolio construction is often treated as a weight-allocation problem over a fixed investable universe. This thesis studies the earlier selection stage: which assets should be considered for an investor before final allocation is applied. The focus is a mixed Egyptian-market universe containing money-market exposure, government bonds, equity exposure, real-estate exposure, and gold.

The thesis proposes an artificial intelligence and machine learning (AI/ML) framework for pre-allocation asset-universe selection based on investor risk tolerance. A point-in-time monthly panel is built from historical daily market data and macroeconomic features. A Proximal Policy Optimization (PPO) reinforcement-learning model assigns each active asset a predicted risk-ranking score, where the target combines realized volatility, downside deviation, and maximum drawdown. Assets are then ranked from low to high predicted risk-ranking score and mapped into conservative, balanced, and aggressive investor-profile universes.

The results show that the promoted PPO risk-ranking model creates historically distinct realized-risk buckets from predicted ranks. On the final test split, the model reaches a mean reward of about 0.75 and a mean Spearman rank correlation of about 0.67. To isolate the effect of the proposed risk-tolerance-based universe construction, the investor-profile universes are compared with a filter-off baseline that keeps the full active universe and applies the same equal-weight historical simulation. Under this comparison, the full active universe has mean realized risk 0.500, while the low-risk and high-risk buckets have 0.239 and 0.688, respectively. The conservative bucket therefore shows materially lower realized risk than the full active universe, while the aggressive bucket shows higher realized risk and higher historical cumulative return in the short final test window. These findings support the thesis claim that asset-universe selection can be treated as a risk-tolerance-oriented stage before allocation, but they do not prove guaranteed outperformance or solve final portfolio-weight optimization.

Contents

Chapter 1

Introduction

1.1 Motivation

Portfolio construction is often presented as an optimization problem over a set of available assets. Classical portfolio theory explains how capital can be distributed once that set is known, but it often leaves an earlier question outside the model: which assets should be considered in the first place? This question matters because the same candidate universe is not suitable for every investor. Before portfolio weights are assigned, the asset universe should be checked against the investor's objectives, risk tolerance, and the current market conditions.

This issue becomes clearer in markets where asset behavior changes over time. In the Egyptian financial market, risk can shift because of inflation, exchange-rate movements, monetary-policy changes, and market stress. In such conditions, treating an asset as permanently conservative or permanently aggressive can be misleading. A portfolio process that starts directly with final allocation may therefore depend on a candidate universe that is already weak for the target investor profile.

For that reason, this thesis treats asset-universe selection as a separate decision stage before allocation. The goal is not to compute final portfolio weights. Instead, the study examines how artificial intelligence and machine learning (AI/ML) can support the selection of an investor-suitable universe before an allocation model is applied. Investor suitability is interpreted through three risk-tolerance groups. Conservative investors prioritize risk control, balanced investors seek a middle risk profile, and aggressive investors accept higher realized-risk exposure. Return behavior is used later only as a historical diagnostic of these risk-tolerance universes.

The implementation studies this problem through an adaptive risk-scoring approach. A reinforcement-learning (RL) model assigns an asset-level predicted risk-ranking score and ranks the active universe in each decision month. The realized-risk target combines realized volatility, downside deviation, and maximum drawdown. These rankings are then used to form risk-tolerance-oriented asset universes before allocation. The contribution is therefore AI-supported asset-risk ranking for investor-suitable universe construction, not final portfolio optimization.

1.2 Problem Statement

Existing portfolio allocation methods commonly assume that the investable universe has already been defined. Related AI and machine-learning preselection studies show that filtering assets before optimization can be studied as its own step. However, these studies commonly select assets using expected return, price movement, profitability, Sharpe-like measures, efficiency, robustness, or final optimizer performance. The problem addressed in this thesis is how to select an investor-suitable asset universe before allocation when suitability is defined by risk tolerance and asset-level realized-risk behavior.

1.3 Scope and Asset Universe

The thesis focuses on a mixed Egyptian-market universe. The investable asset universe includes money-market exposure represented by 91-day Treasury bills, government bonds represented by 5-year bonds, Egyptian equity exposure represented by the Egyptian Exchange 30 (EGX30) index and available EGX30 constituent stocks, real-estate exposure represented by a real estate investment trust (REIT) series, and gold exposure represented by 24K gold in Egyptian pounds.

The current scope ends at pre-allocation universe selection. Final portfolio weighting and investor-facing production serving are not treated as the main implemented contribution. The thesis uses a simple equal-weight historical simulation only as an evaluation diagnostic, so that selected universes can be compared with the full active universe under the same rule.

1.4 Research Questions and Objectives

This thesis is guided by the following research questions:

- RQ1: Can AI/ML support dynamic asset-universe selection before allocation using asset-level realized-risk prediction?
- RQ2: Do the selected universes align with conservative, balanced, and aggressive investor risk-tolerance profiles?
- RQ3: How do the proposed risk-tolerance-based asset universes compare with the full active-universe baseline when each is evaluated under the same equal-weight historical simulation rule?

For RQ3, the empirical baseline is a filter-off ablation. The baseline condition keeps the full active universe, while the proposed condition evaluates the PPO-filtered risk-tolerance universes under the same equal-weight historical simulation rule.

The objectives of the thesis are to:

- design a pre-allocation asset-universe selection framework guided by investor risk tolerance;
- construct point-in-time asset-risk features and composite realized-risk targets;
- train and evaluate a reinforcement-learning model that learns asset-level risk-ranking scores;
- evaluate the selected investor-profile universes against the full active universe using historical realized-risk, return, volatility, and drawdown diagnostics under a common equal-weight construction.

1.5 Outline

The remainder of this thesis is organized as follows. Chapter 2 reviews the relevant literature on portfolio optimization, AI/ML asset preselection, investor-risk personalization, reinforcement-learning methods, and the research gap addressed by this thesis. Chapter 3 defines the methodology, Chapter 4 reports the experimental results, and Chapter 5 presents the conclusion and future work.

Chapter 2

Literature Review

This chapter studies asset-universe selection as a stage that comes before final portfolio weighting. An optimizer can only allocate among assets that are included in the candidate set. If that set contains assets that do not match the investor's risk tolerance, then the allocation stage starts from a weak position. For this reason, the review covers classical portfolio optimization, AI/ML asset preselection, investor-risk personalization, and adaptive reinforcement-learning methods.

The review is organized around one central distinction. Many existing papers show that selecting assets before portfolio optimization can improve later portfolio construction. However, a suitable asset is often defined through expected return, price movement, profitability, Sharpe-like performance, efficiency, robustness, or final optimizer quality. In this work, suitability is defined differently. The focus is whether assets can be selected for conservative, balanced, and aggressive investor groups by directly predicting asset-level realized risk before allocation.

2.1 Portfolio Optimization and the Fixed-Universe Assumption

A classical starting point is the mean-variance framework introduced by Markowitz [1]. The framework formalizes the tradeoff between return and risk when assigning portfolio weights, but it assumes that the investable universe is already known. This assumption is important here because it separates two decisions that are often combined in practice: choosing the candidate assets and assigning final weights.

Later allocation methods provide additional ways to represent investor views and risk objectives. The Black-Litterman model incorporates investor views into allocation decisions [2], while conditional value-at-risk optimization focuses on downside tail risk [3]. Quantile-based portfolio selection also extends the discussion beyond variance by studying distributional risk criteria [4]. These approaches are relevant because they show that risk can be modeled in several ways, but they still mainly operate after the asset universe has been defined.

Benchmarking work also shows why the universe-selection stage should be separated from the weighting stage. DeMiguel et al. [5] showed that naive diversification is a difficult benchmark for optimized portfolios to beat, while hierarchical risk parity later offered a risk-based allocation alternative that emphasizes diversification structure [6]. These methods are useful baselines for allocation, but they do not decide which assets are suitable for a specific investor before weights are computed.

2.2 AI/ML Asset Preselection Before Optimization

The closest literature comparison group to this thesis is AI/ML-based asset preselection before portfolio optimization. Wang et al. [7] proposed a deep-learning preselection framework where selected assets are passed to portfolio formation. This paper is directly relevant because it treats asset filtering as a separate stage before allocation. Its selection logic, however, is not based on direct prediction of an asset-level realized-risk target for investor risk-tolerance grouping.

Similar predict-then-select patterns appear in later studies. Ma et al. [8] connected machine-learning and deep-learning return prediction to portfolio optimization. Kaczmarek and Perez [9] used machine-learning predictions to build portfolios under rules such as mean-variance optimization, hierarchical risk parity, and naive diversification. Chaweewanchon and Chaysiri [10] used predictive stock selection before Markowitz optimization. These studies show that ML can serve as a filter before final allocation, but the filtering signal is mainly driven by return or price prediction.

Other work broadens preselection beyond simple return prediction. Mills and Anyomi [11] proposed a two-stage robust portfolio construction approach under uncertainty. Hosseinzadeh et al. [12] used data envelopment analysis for asset preselection before portfolio optimization. Abdi et al. [13] combined LSTM modeling with Sharpe-ratio maximization for prospective asset preselection. Chou and Pham [14] used ensemble learning for stock preselection with multiobjective investment optimization and investor-preference framing.

Together, these papers show that preselection before allocation is a defensible modeling stage. They also form the main literature comparison family for this thesis, but not a directly matched empirical baseline. The key difference is the definition of suitability. In most of this literature, a suitable asset is one expected to improve return, price prediction, efficiency, Sharpe-like performance, robustness, or optimizer quality. In this thesis, a suitable asset is one whose predicted realized-risk behavior matches the investor's risk-tolerance group before any allocation model is applied.

2.3 Investor Risk Tolerance and Personalization

The preselection literature motivates the stage studied in this thesis, but investor-specific suitability also requires work on personalization. Personalized finance advisory systems show that recommendations can change when investor information is modeled explicitly. Musto et al. [15] used case-based recommendation and diversification strategies for personalized finance advice. Although the output is not a formal asset universe, the paper supports the idea that financial recommendation should not treat all investors as identical.

Recent AI-based personalization studies provide additional support for investor-aware financial decisions. Asemi et al. [16] used an adaptive neuro-fuzzy inference system to customize investment type from investor demographics and feedback. A related model combined ANFIS, clustering, expert feedback, and multimodal neural-network pretraining for investment-type recommendation [17]. Wei and Liu [18] proposed personalized fund recommendation with dynamic utility learning. These papers connect investor information to financial recommendations, but their outputs are usually investment types or fund recommendations rather than an asset-level universe selected before allocation.

Risk tolerance is also studied directly in robo-advising and personalized portfolio design. Alsabah et al. [19] modeled how investor risk preferences can be learned from portfolio choices. Yu and Liu [20] proposed a personalized mean Conditional Value-at-Risk (mean-CVaR) portfolio optimization model for individual investment. Capponi et al. [21] studied personalized robo-advising through client interaction. Schneider and Yilmaz [22] showed that risk appetite can change stock and portfolio recommendations produced by ChatGPT. These studies justify the use of investor risk tolerance as a decision variable, but they do not usually define an explicit pre-allocation asset universe selected through direct asset-risk prediction.

2.4 Adaptive AI and Reinforcement Learning in Finance

Adaptive AI methods are relevant because financial suitability is not static. Deep reinforcement learning has been used for trading and portfolio-management tasks, including financial signal representation and trading decisions [23]. Jiang et al. [24] proposed a deep reinforcement-learning framework for portfolio management, and FinRL later provided environments and tools for automated stock trading and quantitative finance research [25].

Additional deep reinforcement learning (DRL) portfolio studies show how adaptive policies can be evaluated in financial settings. Lucarelli and Borrotti [26] studied deep Q-learning for cryptocurrency portfolio management. Soleymani and Paquet [27] used online deep reinforcement learning with an autoencoder representation for portfolio optimization. Pinelis and Ruppert [28] studied machine-learning portfolio allocation, while

Rezaei and Nezamabadi-Pour [29] provided a taxonomy and experimental study of deep reinforcement learning in portfolio management.

Risk-aware reinforcement learning is especially relevant to the implementation direction of this thesis. Choudhary et al. [30] trained risk-adjusted deep reinforcement-learning agents using reward functions related to return, Sharpe ratio, and maximum drawdown. This supports the use of RL in risk-aware financial modeling. However, the action in such work is still usually a trading or allocation decision, not the prediction of an asset-level risk-ranking score for building a pre-allocation universe.

The closest single comparison found in the current review is the investor-specific DRL framework of Orra et al. [31], which is an arXiv preprint. Their pipeline uses volatility-guided grouping to form conservative, moderate, and aggressive risk profiles before DRL allocation. It is highly relevant because it connects investor risk profiles, asset grouping, and adaptive learning. The remaining difference is that the grouping itself is not learned as an AI/ML asset-universe selection step. In their framework, AI/ML is used for allocation after grouping, while this thesis applies AI/ML to the pre-allocation selection stage by predicting a composite realized-risk target over a mixed and variable asset universe.

Table 2.1 summarizes the main comparison dimensions used to position the thesis. The table groups related papers by methodological family rather than reproducing every reference one by one.

Table 2.1: Literature comparison matrix for the thesis positioning

Literature family	Preselection before allocation	Investor risk profile	Adaptivity and RL use	Asset coverage	Benchmark / evaluation type	Main limitation for this thesis
Classical and risk-based allocation	No	Partial	No RL	Usually fixed universe	Efficient frontier, CVaR, equal weight, risk parity	Strong allocation tools, but the investable universe is usually assumed before weights are assigned.
AI/ML preselection before optimization	Yes	Mostly no	Adaptive ML, usually not RL	Mostly stocks or fixed candidate sets	Optimized portfolio performance, return, Sharpe-like or robustness metrics	Validates preselection, but selection is usually driven by return, price movement, efficiency, or optimizer quality rather than direct risk-tolerance suitability.
Investor-profile and robo-advisory studies	Partial	Yes	Adaptive recommendation in some studies	Funds, products, advice, or final portfolios	Recommendation quality, client utility, or personalized allocation	Models investor differences, but usually does not construct an explicit asset universe before allocation.
RL and DRL portfolio management	Usually no	Mostly no or partial	Yes	Mostly stocks, crypto, or benchmark portfolios	Trading return, allocation performance, drawdown, Sharpe-like diagnostics	Learns trading or allocation actions directly rather than an asset-level risk-ranking stage.
Investor-specific DRL grouping	Yes	Yes	DRL for allocation after grouping	Stock universe in the reviewed preprint	Allocation benchmarks such as equal weight or market indices	Closest comparison, but grouping is volatility-guided rather than an AI/ML risk-ranking model over a mixed variable universe.

Note. The benchmark/evaluation column reports the dominant evaluation focus in each literature family. In recommendation studies, this may be client utility or recommendation quality rather than a directly tradable portfolio benchmark.

2.5 Synthesis and Research Gap

The reviewed literature leads to three observations. First, classical allocation methods provide strong tools for assigning portfolio weights, but they usually assume that the investable universe has already been chosen. Second, AI/ML preselection studies show that selecting assets before optimization can be evaluated separately from weighting. Third, investor-personalization studies show that risk tolerance should affect financial recommendations. The gap appears between these streams rather than inside only one of them.

The matrix shows why no directly matched literature baseline is used as a one-to-one empirical comparator in this thesis. Existing studies solve related parts of the problem, but their datasets, objectives, asset coverage, and selection criteria differ from the implemented task. The main gap is that direct prediction of an asset-level realized-risk ranking for investor-profile universe construction is mostly absent in the reviewed literature. The closest partial exception is Orra et al. [31], but its grouping is volatility-guided rather than an AI/ML pre-allocation selection model and does not predict a composite realized-risk target in the same form used here.

This thesis is positioned as an investor-risk-tolerance-oriented extension of the preselection literature. It studies pre-allocation universe construction by predicting and ranking assets according to a composite `realized_risk` target before final allocation. The selected universes are then interpreted according to conservative, balanced, and aggressive investor profiles. This connects asset preselection, investor risk tolerance, and adaptive AI while keeping final portfolio weighting as a later stage. Chapter 3 translates this gap into the methodology used for the implemented PPO risk-ranking system.

Chapter 3

Methodology

This chapter defines the methodology used in this thesis to construct investor-risk-tolerance-oriented asset universes before final allocation. The motivation is the fixed-universe assumption that appears in many allocation methods: an optimizer can assign weights only after the candidate assets have already been chosen. In this thesis, that earlier universe-construction problem is treated as the main focus. The proposed method estimates asset-level realized-risk behavior, ranks the active universe in each decision month, and maps the ranking to conservative, balanced, and aggressive investor profiles.

The contribution is therefore a pre-allocation risk-ranking framework. Portfolio returns, Sharpe ratios, volatility, and drawdowns are used later as historical diagnostics for selected universes. They are not treated as proof that the model guarantees outperformance or directly optimizes final portfolio weights.

3.1 Methodological Scope and Objective

The methodology addresses the gap identified in Chapter 2: risk is commonly used inside portfolio optimization objectives, constraints, and evaluation metrics, but is less often used as the main signal for investor-specific asset eligibility before allocation. The objective is to estimate which assets are suitable for conservative, balanced, and aggressive investors by learning a month-level asset risk ranking from point-in-time market information.

The framework makes two methodological choices. First, asset suitability is represented through a constructed realized-risk ordering rather than through a return-forecasting target. Second, model development follows a chronological point-in-time protocol: each decision month is scored using only information available before the corresponding realized-risk outcome is evaluated. This design protects against temporal leakage and makes validation and test reporting meaningful as out-of-sample checks.

The chapter therefore moves from data construction to learning and then to evaluation. It first defines the point-in-time monthly panel, then explains how PPO scores the active universe by predicted risk-ranking score, then shows how the ranked assets are mapped to investor-profile buckets and evaluated through historical diagnostics.

3.2 Proposed Framework Overview

Figure 3.1 summarizes the end-to-end methodology. The framework has five layers. The data layer cleans daily market and macroeconomic inputs. The construction layer builds point-in-time monthly asset states and target-only realized-risk fields. The learning layer trains a Proximal Policy Optimization (PPO) risk-ranking policy. The selection layer maps predicted ranks to investor-profile buckets. The evaluation layer compares selected universes with baselines using historical diagnostics.

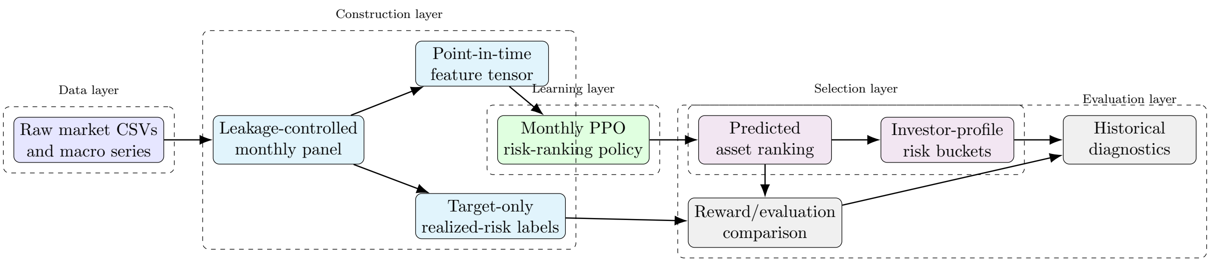

Figure 3.1: Layered methodology pipeline for risk-tolerance-oriented universe construction.

Table 3.1 gives the implementation and promotion roadmap followed by the rest of the chapter. It is a guide to the development sequence rather than a replacement for the detailed methodology sections that follow.

Table 3.1: Methodology implementation and promotion order

Stage	Methodological role and promotion logic
Data and monthly panel construction	Build leakage-controlled point-in-time asset rows and target-only realized-risk fields before any model comparison.
Realized-risk target definition	Use the balanced target based on ranked realized volatility, downside deviation, and maximum drawdown; retain alternatives as target-design audits.
PPO risk-scoring setup	Formulate each decision month as one active-universe ranking problem with a rank-dominant reward and score-discipline term.
Architecture and framework testing	Compare no-context monthly PPO, pooled-context monthly PPO, daily-flat PPO, and daily-CNN PPO under the same validation protocol; promote the monthly pooled-context framework.
Feature-set testing	Test drops, replacements, window changes, and additive candidates after the framework is fixed; retain the broad input families and add the downside-tail contribution ratio.
PPO tuning and final model reporting	Tune PPO with validation reward as the primary criterion and Spearman as a rank-quality guardrail; reserve test results for final reporting.
Investor-profile bucketing and evaluation	Map predicted ranks to conservative, balanced, and aggressive universes, then compare the proposed universes with the full active-universe baseline under the same equal-weight diagnostic rule.

The proposed workflow is evaluated without a user-interaction experiment. Investor profiles are represented by risk-tolerance buckets over the predicted ranking, not by a deployed questionnaire or live user study. This keeps the methodology aligned with the implemented model and avoids claiming a production investor-advisory system.

3.3 Dataset and Point-in-Time Feature Construction

The data engineering pipeline converts daily market and macroeconomic data into a monthly long panel. One row represents one active (month, asset) state. The canonical shape is long rather than wide because the active universe changes through time; an asset appears only after real source history exists.

Figure 3.2 shows how raw inputs are split into metadata, model-input features, and target-only fields. Table 3.2 lists the data sources and their methodological roles.

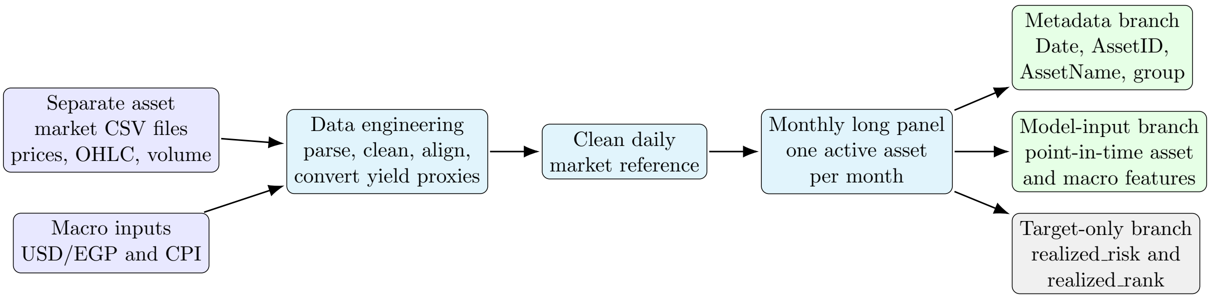

Figure 3.2: Data pipeline from separate source files to metadata, model-input, and target-only monthly panel branches.

Table 3.2: Raw data sources used by the pipeline

Series	Data type	Methodological role
91-day treasury bills	Market yield series	Low-risk money-market exposure; yield converted to price proxy for returns.
5-year government bonds	Market yield series	Government fixed-income exposure; yield converted to price proxy for returns.
EGX30 index	Market index series	Equity-market benchmark and beta reference.
EGX30 constituents	Daily equity market CSVs	Expanded equity universe; rows begin only when real history exists.
REIT index	Market index series	Real-estate exposure proxy.
Gold	Market price series	Gold exposure in Egyptian pounds.
USD/EGP exchange rate	Macro feature	Foreign-exchange context after concatenation and deduplication.
CPI	Macro feature	Inflation context; monthly file cleaned separately from daily market data.

Raw market CSV files use a tabular market-data format with dates, price, open, high, low, volume, and quoted percentage change. Numeric fields are parsed after removing formatting characters; volume suffixes are converted into numeric values. The vendor change field is retained as a quality-control field, while authoritative returns are computed from cleaned prices. Money-market and bond series are quoted as yields, so fixed-maturity price proxies are used before return and range-derived features are calculated.

Daily series are aligned to the Egyptian market calendar. Short forward filling is applied only to quoted level series where a missing observation reflects non-trading or publication timing rather than a missing asset history. Audit fields, range-price fields, volume, and vendor change values remain missing on synthetic forward-filled rows. Pre-listing history is never imputed; an absent row means the asset is unavailable for that month.

Table 3.3 groups the model inputs by financial role. The volatility family includes walk-forward EGARCH summaries, following Nelson's conditional heteroskedasticity specification [32]. The model-facing tensor excludes `AssetID`, `AssetName`, and `AssetGroup`; those identifiers are retained only as metadata.

Table 3.3: Feature families used for point-in-time asset risk scoring

Feature family	Methodological role
Volatility and risk	Captures recent realized variation, walk-forward EGARCH volatility summaries, drawdown behavior, and downside variation.
Downside and tail behavior	Captures harmful return asymmetry and recent downside-tail contribution.
Liquidity	Represents trading activity through observed volume where available.
Momentum and technical state	Captures relative price location, range behavior, momentum, and overbought or oversold state.
Market beta	Measures recent co-movement with the EGX30 benchmark.
Macro context	Represents USD/EGP foreign-exchange volatility and CPI inflation trajectory shared across the active universe.

The implemented feature names and the final promoted feature set are reported in Chapter 4. Algorithm 1 summarizes the panel-construction process used by the methodology.

Algorithm 1 Point-in-time monthly panel construction

1: Parse market and macro source files into cleaned daily series.
2: Convert yield-quoted fixed-income series into price proxies before return calculation.
3: Align daily observations to the Egyptian market calendar and apply limited level-only forward filling.
4: For each month, construct active asset rows only when real historical source data exists.
5: Compute point-in-time asset features from trailing information available by the state month.
6: Repeat macro features across active assets in the same month.
7: Compute target-only realized-risk components from realized returns inside the target month.
8: Store metadata, model-input features, and target-only fields in a long monthly panel.

3.4 Composite Realized-Risk Target

The realized-risk target defines the cross-sectional ordering that the PPO model learns to recover. The thesis does not depend on interpreting the exact score as a calibrated risk forecast. Instead, the central learning problem is whether the model can recover the low-to-high ordering induced by realized volatility, downside deviation, and maximum drawdown.

These components are chosen because each captures a different form of risk exposure: volatility measures overall return variation, downside deviation focuses on negative variation, and maximum drawdown captures peak-to-trough loss severity.

For component k in decision month t, $x_{i,t}^{(k)}$ denotes asset i's realized component value, such as realized volatility, downside deviation, or maximum drawdown. The operator $\mathrm{rank}_t(\cdot)$ ranks only the assets active in the same month from lower realized risk to higher realized risk. Equation 3.1 converts that within-month component rank into a normalized rank $q_{i,t}^{(k)}$. Months with too few active assets are removed before modeling.

$$q_{i,t}^{(k)} = \frac{\mathrm{rank}_t(x_{i,t}^{(k)}) - 1}{N_t - 1}, \tag{3.1}$$

Here, N_t is the number of active assets in month t. A $q_{i,t}^{(k)}$ value near 0 means the asset is near the low-risk end of that component for the month, while a value near 1 means it is near the high-risk end. The promoted realized-risk target $s_{i,t}$ is the equal-weighted average of the normalized volatility, downside, and drawdown ranks, as shown in Equation 3.2.

$$s_{i,t} = \frac{1}{3} q_{i,t}^{(\mathrm{vol})} + \frac{1}{3} q_{i,t}^{(\mathrm{downside})} + \frac{1}{3} q_{i,t}^{(\mathrm{drawdown})}. \tag{3.2}$$

This balanced target is promoted as a methodological choice rather than as a test-selected target. No single risk concept dominates, the components are scale-free within each month, and the target matches the objective of investor suitability by risk rather than return prediction. Table 3.4 lists sensitivity definitions considered during development. These alternatives are retained as target-design audits and were not promoted as the final realized-risk target.

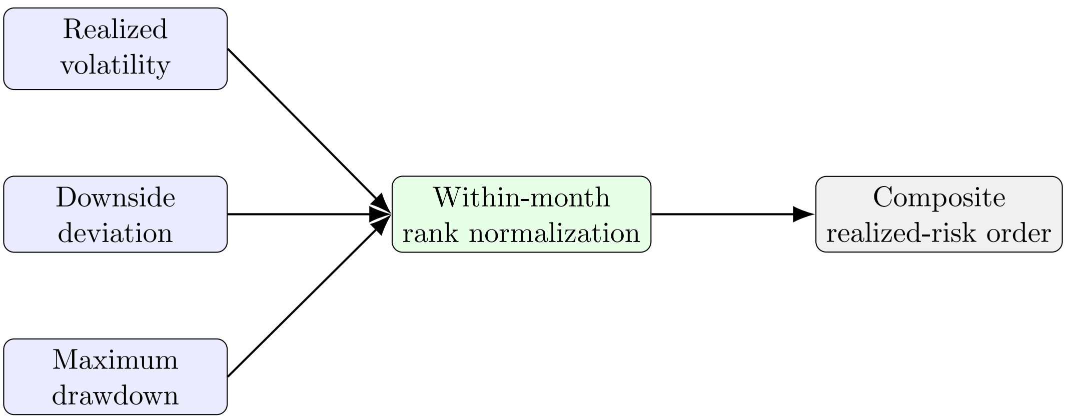

Figure 3.3: Composite realized-risk target construction.

Table 3.4: Realized-risk objective profiles considered during development

Target definition	Vol.	Downside	Drawdown	Purpose
Balanced risk definition	1/3	1/3	1/3	Promoted balanced risk definition.
Downside-heavy definition	0.25	0.50	0.25	Downside-heavy sensitivity definition.
Tail-and-drawdown definition	0.20	0.40	0.40	Tail and drawdown-heavy sensitivity definition.
Volatility-only definition	1.00	0.00	0.00	Volatility-only stress case.
Downside-only definition	0.00	1.00	0.00	Downside-only stress case.
Drawdown-only definition	0.00	0.00	1.00	Drawdown-only stress case.
Downside-and-drawdown definition	0.00	0.50	0.50	Downside and drawdown-only sensitivity definition.

3.5 Reinforcement-Learning-Based Risk Scoring

The active implementation uses PPO, a policy-gradient reinforcement-learning algorithm designed to update policies through a clipped objective that limits destructive policy changes [33]. PPO is used here because the task is naturally expressed as repeated month-level ranking over a changing active universe. Each decision month contains a different set of available assets, a different market context, and a different realized-risk ordering.

One PPO episode corresponds to one decision month. The observation is a padded feature tensor and an active-asset mask. The action is one bounded continuous risk-ranking score for each padded asset slot. Padded rows are retained only to keep tensor shapes fixed; they do not contribute to action sampling, log probability, entropy, loss terms, reward, or evaluation metrics.

Assets are sorted from low to high predicted risk-ranking score to form the predicted rank order. The promoted reward is computed once per month across the active universe and combines Spearman rank correlation with a mean squared error (MSE) score term, as shown in Equation 3.3.

$$R_t = 0.7 \cdot \rho_s(\hat{r}_t, r_t) + 0.3 \cdot (1 - \mathrm{MSE}(\hat{s}_t, s_t)), \tag{3.3}$$

where ρ_s is the Spearman rank correlation between predicted and realized ranks, $\hat{s}_t$ is the predicted score vector, and s_t is the realized-risk target vector. The Spearman term reflects the primary selection objective: recovering the cross-sectional ordering. The MSE term discourages arbitrary score scaling while remaining secondary to ranking quality. Table 3.5 records the reward definitions considered during development.

Table 3.5: Reward profiles considered during PPO development

Reward definition	Spearman	$1 - \text{MSE}$	Tail check	Methodological purpose
Rank-dominant reward	0.70	0.30	0.00	Promoted reward; rank-dominant with score discipline.
Stronger-rank reward	0.85	0.15	0.00	Stronger rank emphasis.
Pure-rank reward	1.00	0.00	0.00	Pure rank-only audit.
Balanced-rank-score reward	0.50	0.50	0.00	Balanced rank and score audit.
Tail-aware reward audit	0.60	0.20	0.20	Audit for whether extra tail emphasis helped identify high-realized-risk assets.

Algorithm 2 summarizes the monthly scoring, ranking, and bucketing process.

Algorithm 2 Monthly PPO risk scoring and investor-bucket construction

1: Select a decision month t and assemble prior monthly state rows required by the active framework.
2: Build the padded feature tensor and active-asset mask.
3: Encode active asset rows with the shared row encoder.
4: Build a mask-aware monthly context from active row representations using mean pooling, max pooling, and active-asset percentage.
5: Pass the same monthly context to each active asset representation.
6: Apply the PPO actor to generate one bounded predicted risk-ranking score per active asset.
7: Sort active assets from low to high predicted risk-ranking score.
8: Convert each sorted position into a predicted-rank percentile.
9: Assign assets to investor-profile buckets using rank-percentile bands.
10: During training and evaluation, compare predicted ranks with target-only realized-risk ranks.
11: Report selected-universe diagnostics separately from any final allocation method.

3.6 Actor-Critic Architecture

The actor-critic architecture is the model structure used by PPO to learn from monthly ranking decisions. For each decision month, the policy observes the active universe as a masked table: each row represents one available asset, each column represents a point-in-time feature, and the mask identifies which rows are real assets rather than padding. The policy must turn this table into a set of predicted risk-ranking scores that can be sorted into an asset ranking.

The actor is the scoring side of the policy. It reads each active asset row and assigns a bounded predicted risk-ranking score, so its output directly determines the low-to-high ranking used for investor-profile selection. The critic is the evaluation side of the policy. It receives the pooled monthly context and estimates the expected PPO reward for the current month's full ranking decision before the realized reward is observed. PPO compares the actual monthly reward with this estimate to form the advantage signal used to update the actor.

Both sides begin from the same row encoder. In the implemented model, this row encoder is a multilayer perceptron that converts each asset's feature row into a 64-number representation. Conceptually, this applies the same risk-reading rule to every asset row, so equities, gold, bonds, and money-market exposure are processed by a common feature-based scorer rather than by separate asset-specific models. A pooled context vector then summarizes the active universe in that month using the mean, maximum, and active-asset percentage of the available row representations. This context is passed to each active asset representation before scoring, allowing the actor to judge an asset relative to the other assets available at the same decision date. The actor and critic are separate multilayer perceptron heads: the actor produces asset-level predicted risk-ranking scores, while the critic produces one expected-reward estimate for the monthly ranking decision. The mask keeps unavailable or padded rows invisible to scoring, value estimation, learning, reward, and evaluation.

Figure 3.4 illustrates the architecture, and Table 3.6 defines its main components.

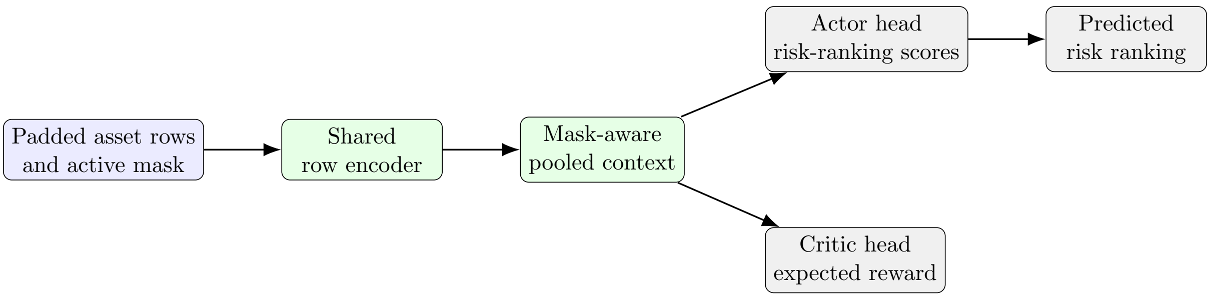

Figure 3.4: Masked PPO actor-critic architecture for variable active universes.

Table 3.6: Main actor-critic architecture components

Component	Methodological role
Input tensor	Padded active-universe feature matrix plus an active-asset mask.
Shared row encoder	A multilayer perceptron applied to every asset row with the same parameters, producing a 64-number row representation.
Pooled context	Mask-aware active-universe summary using mean pooling, max pooling, and active-asset percentage.
Actor head	Asset-level multilayer perceptron head that receives the row representation and pooled context, then outputs one bounded predicted risk-ranking score distribution per active asset.
Critic head	Month-level multilayer perceptron head that receives the pooled context and estimates the expected PPO reward for the full monthly ranking decision.
Action distribution	Masked sigmoid-squashed Gaussian distribution, so risk-score actions are bounded in $[0, 1]$.
Masking behavior	Padded rows are excluded from sampling, log-probability, losses, reward, and metrics.

Daily-input variants were evaluated because month-end features may hide short-lived daily patterns such as abrupt selloffs, volume shocks, or local momentum reversals. The daily-flat variant concatenated recent observed daily values into the asset-row scorer, while the daily-strip convolutional neural network (CNN) variant encoded a short prior-month daily strip. These paths were useful development checks but were not promoted, because they did not improve validation ranking quality over the monthly pooled-context design.

3.7 Method Selection and Promotion Protocol

The promotion protocol answers how candidate options were compared before the final model was reported. The development process used controlled experiments: within each comparison stage, the tested variable changed while the other major design choices were held fixed. This made architectural, feature, reward, target, bucketing, and tuning decisions comparable without mixing several changes into the same result.

Validation reward was the primary promotion metric because it is the model's training objective and combines rank alignment with score discipline. Spearman rank correlation was retained as a guardrail and diagnostic rather than as a separate co-primary objective, because the promoted reward is itself rank-dominant. Test results were not used to choose among candidate options; they were reserved for final reporting in Chapter 4.

The framework comparison asked whether the policy should score assets independently, use the pooled active-universe context described in Section 3.6, or add recent daily paths through flat or CNN encoders. The promoted representation is the monthly pooled-context framework; daily-flat and daily-CNN variants were compared as candidate families under the same validation protocol.

Table 3.7 summarizes the model families considered. Final numerical results for the promoted framework are reported in Chapter 4.

Table 3.7: Framework and model-family decisions

Candidate family	Role considered	Methodological decision
Monthly PPO without pooled context	Scores each asset using its own point-in-time features, without a pooled summary of the active universe.	Useful baseline, but less effective than the contextual monthly design.
Monthly PPO with pooled context	Scores each asset using point-in-time features plus a pooled active-universe summary.	Promoted risk-ranking representation.
Daily-flat PPO	Adds a flattened recent daily path to the asset scorer.	Evaluated and rejected; did not improve the monthly pooled-context system.
Daily-CNN PPO	Encodes a short observed daily strip with convolutional filters.	Evaluated and rejected; useful audit for local daily patterns.

3.8 Feature-Set Testing and Input Selection

Feature-set testing began after the monthly pooled-context framework had been selected. This ordering was important because feature effects should be judged inside the same model structure that would later be tuned and reported. The promotion rule matched the framework-selection rule: each candidate was compared under controlled conditions, validation reward was the primary metric, Spearman rank correlation was retained as the rank-quality guardrail, and the test split was reserved for final reporting.

The first feature step defined the baseline feature families. The model used risk and volatility features, downside-risk features, liquidity features, technical and momentum features, market-sensitivity features, and macroeconomic context features. This gave the feature phase a broad starting point: the candidate set could be tested as a risk-ranking input design rather than as isolated univariate predictors.

The next group of checks used leave-one-out and feature-drop tests. These experiments asked whether a feature family could be removed without weakening validation ranking quality. The purpose was diagnostic: if removing a field improved validation reward or preserved reward with simpler inputs, that feature family would be considered for removal or redesign. The drop checks were therefore used to identify weak or redundant inputs before adding new information.

Replacement and window checks then tested whether selected features should use shorter, longer, or differently defined windows. Examples included replacing the three-month high-distance signal with a one-month high-distance signal, testing a 14-day moving-average distance instead of a 20-day moving-average distance, using one-month maximum drawdown, changing USD volatility to a shorter window, and checking market-beta alternatives. These comparisons kept the broad feature family fixed while changing the definition of the candidate input.

The additive tail and ratio checks tested whether the selected feature set needed an extra signal for abrupt downside behavior. These candidates included downside-adjusted ratios, drawdown-aware ratios, expected shortfall, drawdown duration, worst-return measures, absolute-return shock measures, volatility instability, and downside-tail contribution. The main feature-profile phase was completed before PPO hyperparameter tuning. PPO tuning then fixed the final training configuration as the final model-development step.

The final selected input set kept the broad baseline families and added the three-month downside-tail contribution ratio. Chapter 4 reports the descriptive final feature list, feature-selection evidence, and tuned PPO configuration.

3.9 Investor Risk Profile Mapping and Bucketing

After PPO inference, active assets are ranked from low to high predicted risk-ranking score. The ranked universe is then mapped to investor profiles using rank-percentile bands. These percentiles describe the share of assets after sorting, not the raw score value.

The mapping is rank-based. After assets are sorted by predicted risk-ranking score, each asset's sorted position is converted into a percentile between 0 and 1. The lowest ranked asset is placed near 0, the highest ranked asset is placed near 1, and each investor-profile bucket includes the assets whose percentile falls inside that bucket's range.

Table 3.8 reports the bucket-range candidates considered during development. Chapter 4 reports the numerical evidence for the reported diagnostic rule.

Table 3.8: Investor-profile bucket ranges considered during development

Method	Conservative	Balanced	Aggressive	Methodological decision
Broad overlapping buckets	0.00–0.40	0.25–0.75	0.60–1.00	Broad overlapping audit with moderate tail selectivity.
Non-overlapping thirds	0.00–0.33	0.33–0.67	0.67–1.00	Non-overlapping tercile audit for a simple mutually exclusive split.
Wide overlapping buckets	0.00–0.50	0.20–0.80	0.50–1.00	Wider participation audit with softer separation between profiles.
Selective overlapping-tail buckets	0.00–0.30	0.20–0.80	0.70–1.00	Reported diagnostic rule; selective tails with a broad balanced middle profile.

Bucket diagnostics use realized-risk separation as the main criterion. The primary comparison metric is the high-minus-low mean realized-risk spread between the aggressive and conservative buckets. Supporting checks require the mean realized-risk ordering to increase from conservative to balanced to aggressive, monthly monotonicity to remain high, conservative-bucket realized risk to sit below the full active universe, and aggressive-bucket realized risk to sit above the full active universe. Return diagnostics are reported later as economic interpretation, while bucket evidence is interpreted through risk separation.

Overlap is intentional because investor suitability is not always mutually exclusive. A relatively stable asset can be acceptable for both conservative and balanced investors, while an upper-middle-risk asset can be acceptable for balanced and aggressive investors. Figure 3.5 illustrates this conceptual overlap.

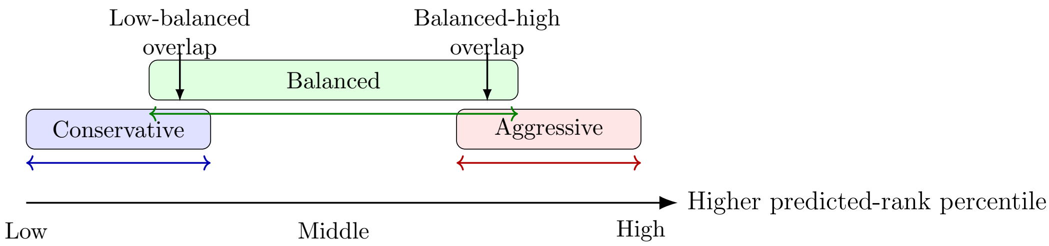

Figure 3.5: Overlapping investor risk buckets on the predicted asset ranking.

3.10 Evaluation Protocol

Evaluation focuses first on model ranking quality. The primary model check compares the predicted asset ranking against the realized-risk ranking for the same decision month. This verifies whether the PPO scorer produces a useful risk order for the active universe.

The second layer evaluates selected investor-profile universes against the full active universe. This comparison isolates the universe filter by holding the diagnostic simulation rule constant. In the implemented results, the full active universe and the selected profile universe are both passed through the same equal-weight historical simulation. The resulting historical outputs are then compared with return and risk diagnostics on the held-out test split.

Figure 3.6 shows the evaluation layers used in this thesis.

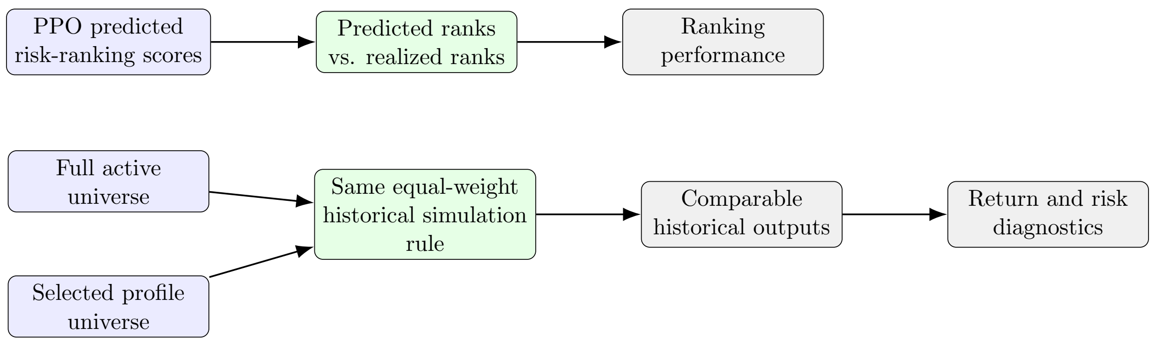

Figure 3.6: Evaluation logic for ranking quality and selected-universe diagnostics before final allocation.

The baseline design uses an internal filter-off ablation motivated by the literature gap in Chapter 2. The empirical baseline disables PPO filtering and evaluates the full active universe under the same equal-weight diagnostic rule used for the filtered investor-profile universes. This keeps the comparison focused on risk-tolerance-based universe construction rather than on a separate optimizer.

3.11 Chapter Summary

This chapter defined the thesis methodology for risk-tolerance-oriented asset-universe construction. It described the point-in-time data panel, realized-risk target, PPO actor-critic scorer, bucket mapping, and evaluation protocol while keeping allocation as a downstream layer. Chapter 4 reports the selected configuration, ranking results, bucket diagnostics, baseline comparisons, and limitations.

Chapter 4

Results

This chapter reports the experimental evidence for the methodology introduced in Chapter 3. The results are organized around the three research questions: whether the model supports dynamic realized-risk ranking, whether the selected universes align with investor risk profiles, and how the selected universes compare with the full active-universe baseline under the same equal-weight diagnostic rule. The chapter reports historical diagnostics only; it does not claim guaranteed future outperformance.

4.1 Experimental Environment

The experiments were run in the local project environment associated with the PPO risk-ranking repository. Table 4.1 lists the verified software environment used for final reporting. The same repository contains the data-processing scripts, PPO training code, evaluation utilities, and final model files.

Table 4.1: Experimental environment used for reporting

Item	Value
Operating environment	Local Windows workstation using PowerShell and the project virtual environment.
Python	3.13.1
Core packages	pandas 3.0.1, NumPy 2.4.3, SciPy 1.17.1, Matplotlib 3.10.8.
Econometric and ML packages	arch 8.0.0, Stable-Baselines3 2.7.1, Gymnasium 1.2.3, Optuna 4.8.0.

4.2 Data Coverage and Splits

The active model uses the monthly long panel described in Section 3.3. The generated panel covers October 2010 through January 2026. Decision months are chronological, and the test period is used only after model, feature, and tuning decisions have been made.

Table 4.2 reports the split design. The training window is used for model learning, inner validation is used for checkpoint selection inside a run, validation is used for framework and candidate comparison, and test is reserved for final reporting.

Table 4.2: Chronological data splits used in the promoted evaluation

Split	Decision months	Months	Methodological role
Training	2011-01 to 2021-12	126	PPO parameter learning.
Inner validation	2022-01 to 2022-12	12	Best checkpoint selection within the final run.
Validation	2023-01 to 2025-02	26	Framework, feature, tuning, and candidate comparison.
Test	2025-03 to 2026-01	11	Final reporting only.

4.3 Model and Parameter Selection

The model-selection process followed the promotion protocol in Table 3.1. First, candidate input frameworks were compared while keeping the initial PPO setup fixed. The monthly framework with pooled active-universe context was selected because it produced the strongest validation reward while preserving the rank-alignment diagnostic. Table 4.3 summarizes the key framework-selection evidence using thesis-readable framework labels.

Table 4.3: Framework-selection evidence before feature and PPO tuning

Framework or check	Validation reward	Spearman diagnostic	Interpretation
Monthly PPO without pooled context	0.6710	0.5608	Strong monthly baseline without active-universe context.
Monthly PPO with pooled context	0.6845	0.5761	Selected framework; pooled active-universe context improved validation ranking quality.
Daily-flat PPO	0.5572	0.3962	Rejected; flattened daily path did not improve ranking quality.
Daily-CNN PPO	0.6650	0.5486	Rejected; strongest daily-input follow-up remained below the selected monthly framework.

After the architecture was fixed, the feature phase tested whether the model inputs should be removed, redefined, shortened, or expanded. The main feature-profile phase preceded PPO hyperparameter tuning and used the same validation rule as the framework comparison. PPO tuning was then treated as the final model-development step. Table 4.4 summarizes what each feature-testing phase changed, retained, or rejected.

Table 4.4: Feature-selection evidence before final reporting

Testing phase	What was tested	Feature-selection outcome
Baseline family screen	Risk, downside, liquidity, technical, market-sensitivity, and macro context features were tested as the starting input family after the framework was fixed.	Retained the broad baseline feature families as the core input design.
Leave-one-out and drop checks	Individual feature families were neutralized or removed to test whether they weakened or improved validation ranking quality.	Tested removals, including the high-distance feature, but no dropped-feature profile was retained in the final current-best input set.
Replacement and window checks	Shorter or alternative definitions were tested for selected inputs, including high-distance, moving-average, drawdown, USD-volatility, and beta variants.	Kept the canonical definitions where alternatives did not improve validation ranking quality.
Tail and ratio additions	Downside, tail, jump, drawdown-duration, ratio, and volatility-instability additions were checked to improve high-risk identification.	Added the three-month downside-tail contribution ratio to the final promoted input set.

Table 4.5 lists the final model inputs using descriptive labels. Most asset-level inputs use the trailing three-month feature window, while some technical indicators use their own observation count inside that window. Macro inputs are repeated for every active asset in the same decision month.

Table 4.5: Final promoted model input features

Input feature	Duration or window	Description
Conditional volatility	Trailing 3 months	Walk-forward EGARCH volatility summary for recent asset return variation.
Downside deviation	Trailing 3 months	Downside-risk measure based on negative daily returns in the recent window.
Maximum drawdown	Trailing 3 months	Largest peak-to-trough loss observed in the recent return path.
Trading volume	Trailing 3 months	Observed trading activity, used as a liquidity and market-attention input.
Average true range percentage	20 observations	Range-based volatility measure divided by the latest observed close.
Market beta	Trailing 3 months	Sensitivity of the asset's returns to the EGX30 reference series.
Price versus moving average	20 observations	Latest price position relative to a simple moving-average benchmark.
Relative strength index	14 observations	Momentum and overbought-or-oversold state based on recent price movement.
Distance from recent high	Trailing 3 months	Latest price distance from the highest observed price in the recent window.
USD exchange-rate volatility	Trailing 3 months	Foreign-exchange volatility context shared across active assets in a month.
CPI trajectory	Trailing 3 months	Inflation trajectory context shared across active assets in a month.
Downside-tail contribution ratio	Trailing 3 months	Share of total absolute movement explained by the largest downside daily-return tail observations.

The final promoted model uses the selected monthly pooled-context framework, the reported input feature set, the promoted realized-risk target, the promoted reward design, and the tuned PPO configuration. Table 4.6 summarizes the evaluated configuration without repeating the full feature list or the tuned hyperparameter values.

Table 4.6: Promoted model configuration

Item	Promoted setting
Promoted model	PPO risk-ranking model.
Framework	Monthly PPO with pooled active-universe context.
Input features	Final promoted feature set listed in Table 4.5.
Realized-risk target	Equal-ranked realized volatility, downside deviation, and maximum drawdown.
Reward profile	70% Spearman rank correlation and 30% score-error discipline.
Training procedure	Chronological ordered-cycle PPO training over the training months.

The alternative realized-risk target and reward definitions described in the methodology are treated here as development audits. The remaining results evaluate the promoted configuration summarized in Table 4.6, rather than adding unreported target-weight or reward-weight comparison tables.

Table 4.7 gives the final tuned PPO hyperparameters, including the Generalized Advantage Estimation (GAE) smoothing parameter.

Table 4.7: Tuned PPO hyperparameter configuration

Hyperparameter	Value
Learning rate	2.4935×10^{-4}
`n_steps`	256
`batch_size`	256
`n_epochs`	10
γ	1.0
GAE λ	1.0
Clip range	0.2990
Entropy coefficient	0.00235
Value-function coefficient	0.9024
Maximum gradient norm	0.3

4.4 Ranking Results

The first research question asks whether AI/ML can support dynamic asset-universe selection using asset-level realized-risk ranking. For this question, the main evidence is the PPO reward because it is the objective used to compare ranking candidates. Spearman rank correlation is also reported because it explains the rank-alignment component inside the reward.

Table 4.8 reports reward first, followed by the rank-alignment diagnostic. The promoted model reached a mean test reward of about 0.75. Because the reward is rank-dominant, the positive Spearman values show that the model's reward aligns with rank ordering rather than score scaling alone.

Table 4.8: Promoted model ranking metrics by split

Split	Months	Mean assets	Reward	Spearman diagnostic
Train	126	29.48	0.7452	0.6574
Inner validation	12	35.75	0.7233	0.6270
Validation	26	36.00	0.7108	0.6081
Test	11	36.00	0.7545	0.6690

Figure 4.1 visualizes the final test split by comparing predicted rank percentiles with realized-risk rank percentiles. Points near the diagonal represent assets whose predicted rank was close to the realized-risk rank in their decision month.

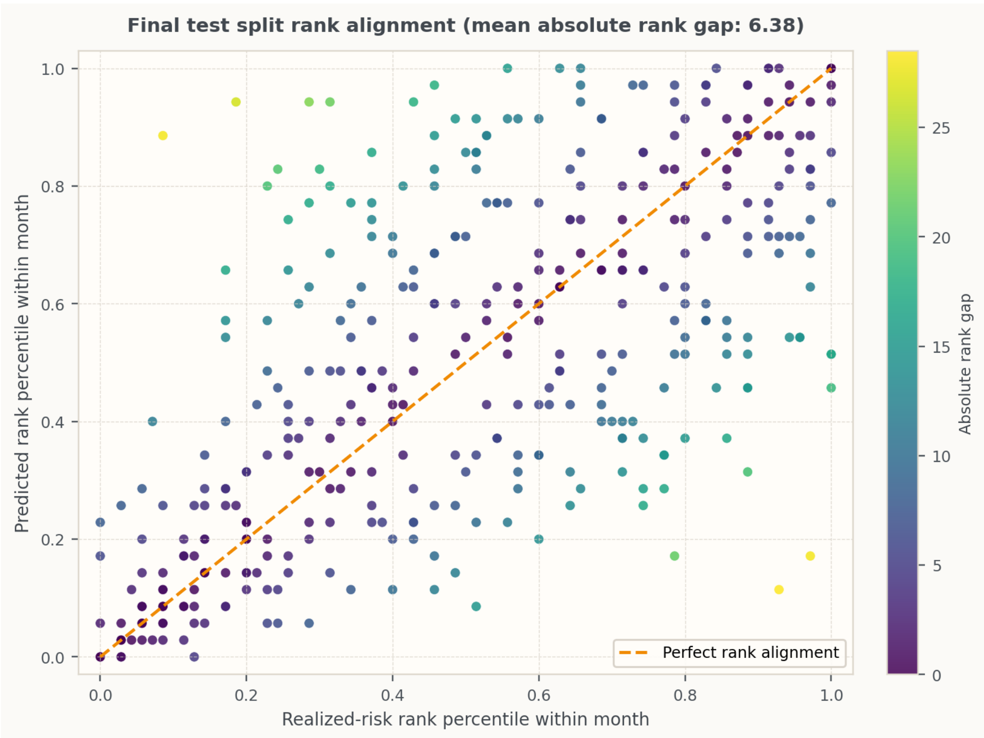

Figure 4.1: Predicted rank percentile versus realized-risk rank percentile on the final test split.

The final selected setup was also checked across multiple random seeds before reporting. The validation reward values from that check were close to the final reported metrics in Table 4.8, while the Spearman diagnostic remained positive. This supports the stability of the selected setup without changing the final test interpretation.

Table 4.9 reports the monthly test-window ranking metrics. The model does not perform equally in every month. October 2025 is the weakest test month by reward, but the reward remains positive across all test months.

Table 4.9: Monthly ranking metrics on the final test split

Month	Reward	Spearman diagnostic
2025-03	0.7786	0.7017
2025-04	0.7702	0.6895
2025-05	0.7456	0.6548
2025-06	0.7019	0.5955
2025-07	0.8033	0.7367
2025-08	0.8270	0.7727
2025-09	0.8088	0.7473
2025-10	0.6399	0.5095
2025-11	0.7595	0.6746
2025-12	0.7548	0.6714
2026-01	0.7102	0.6053

4.5 Risk-Bucket Diagnostics

The second research question asks whether the selected universes align with conservative, balanced, and aggressive investor profiles. The reported bucket method follows the selective overlapping-tail rule defined in Section 3.9: the conservative bucket uses the lowest predicted-rank tail, the balanced bucket uses the broader middle band, and the aggressive bucket uses the highest predicted-rank tail.

Table 4.10 reports the primary bucket evidence. Mean predicted risk-ranking score and mean realized risk are both monotonic from low to medium to high. Because realized risk is rank-normalized within each month, these values should be interpreted as relative separation diagnostics rather than absolute risk magnitudes. The realized-risk difference column is computed against the full active-universe reference of 0.500. The low-risk bucket is materially below that reference, while the high-risk bucket is above it. Monthly realized-risk monotonicity holds in all 11 test months.

Table 4.10: Risk separation for the final test split

Bucket	Mean assets	Mean predicted risk-ranking score	Mean realized risk	Realized-risk difference vs full universe
Low-risk bucket	11	0.406	0.239	-0.261
Medium-risk bucket	22	0.526	0.536	0.036
High-risk bucket	11	0.617	0.688	0.188

Table 4.11 reports secondary economic diagnostics. The high-risk bucket has the highest cumulative return in the short test window, while also having the highest volatility and the largest drawdown among the selected buckets. This is consistent with a risk-return gradient and should be read as a historical diagnostic rather than a return-prediction claim.

Table 4.11: Secondary economic diagnostics for the final test split

Bucket	Cumulative return	Mean monthly return	Monthly volatility	Max drawdown	Sharpe ratio (zero risk-free rate)
Full universe	49.59%	3.75%	1.84%	0.00%	8.673
Low-risk bucket	29.91%	2.43%	1.93%	-1.92%	4.943
Medium-risk bucket	50.17%	3.79%	2.08%	0.00%	7.753
High-risk bucket	86.24%	5.92%	4.71%	-3.97%	5.956

4.6 Filter-Ablation Baseline Comparison

The third research question compares the proposed risk-tolerance-based asset universes with the full active-universe baseline. As established in Chapter 2, related AI/ML pre-selection studies remain comparison context rather than one-to-one empirical baselines. The implemented baseline is a filter-off ablation: the full active universe is evaluated without PPO filtering, and the filtered investor-profile universes are evaluated after PPO filtering.

The empirical baseline keeps the weighting rule fixed and compares equal-weight results for the full active universe against equal-weight results for the filtered investor-profile universes. This isolates the risk-tolerance universe construction: the question is whether the assets selected by predicted rank percentile behave differently from the full universe under the same simple weighting rule. Table 4.12 reports the final test split diagnostics.

Table 4.12: Equal-weight full-universe and filtered-universe diagnostics on the final test split

Universe	Mean realized risk	Cumulative return	Monthly volatility	Max drawdown
Full universe equal weight	0.500	49.59%	1.84%	0.00%
Low-risk filtered equal weight	0.239	29.91%	1.93%	-1.92%
Medium-risk filtered equal weight	0.536	50.17%	2.08%	0.00%
High-risk filtered equal weight	0.688	86.24%	4.71%	-3.97%

The low-risk filtered universe has much lower realized risk than the full universe, while the high-risk filtered universe has higher realized risk and higher cumulative return in this short historical window. The comparison is therefore evidence that the filter changes the risk character of the selected universe under a common equal-weight rule, rather than evidence that a final allocation method is superior.

Validation bucket diagnostics support the same interpretation but also show that the method is not perfect. For the reported overlapping-tail method, validation mean realized risk is ordered from low to medium to high, and monthly monotonicity holds in 24 of 26 validation months. The exceptions are April 2023 and December 2023, where the high-risk bucket remains above the low-risk bucket but falls slightly below the medium bucket.

Table 4.13 reports the final-test bucket-method ablation as sensitivity evidence. The reported overlapping-tail rule shows the largest high-minus-low realized-risk spread and maintains monthly monotonicity across all test months, while wider buckets are easier to populate but create softer separation. This table is not used to turn the test period into a model-selection input.

Table 4.13: Bucket-method ablation on the final test split

Method	Monthly monotonicity	Low risk	Medium risk	High risk	High-low spread
Selective overlapping tails	11/11	0.239	0.536	0.688	0.449
Non-overlapping thirds	10/11	0.257	0.559	0.683	0.426
Broad overlapping tails	11/11	0.304	0.549	0.660	0.356
Wide overlapping buckets	11/11	0.341	0.536	0.659	0.317

4.7 Discussion and Limitations

The results support RQ1 because the promoted PPO model produces positive validation and test reward over a changing active universe. The final test reward of 0.7545 summarizes the rank-dominant objective, while the Spearman diagnostic confirms alignment between predicted and realized-risk ordering. The monthly table also shows month-to-month variation.

The results support RQ2 because the selected buckets create distinct realized-risk groups. The conservative bucket is materially below the full universe, and the aggressive bucket is materially above it. The balanced bucket is not designed to minimize risk; it is intentionally broad and overlapping, so its role is closer to a central participation universe than a low-risk universe.

The results address RQ3 through the filter-off baseline comparison. Under the same equal-weight diagnostic rule, the PPO-filtered conservative universe has lower realized risk than the full active universe, while the PPO-filtered aggressive universe has higher realized risk and higher historical cumulative return in this short test window. This shows that the proposed risk-tolerance-based asset universes have different historical risk characteristics from the full active universe before downstream allocation. The literature comparison remains conceptual context rather than a directly matched empirical baseline.

There are several limitations. The final test window contains only 11 months and is strongly positive for the full active universe. The high-risk bucket's high cumulative return is a diagnostic observation rather than evidence of return prediction. The methodology does not implement production questionnaire inference or user-interaction evaluation, and it makes no claim that the selected universes will improve every final allocation method. The safest conclusion is narrower: the promoted model creates historically distinct realized-risk buckets from predicted ranks.

Chapter 5

Conclusion & Future Work

This chapter concludes the thesis by summarizing the evidence for the research questions, clarifying the contribution of the proposed pre-allocation risk-ranking framework, and outlining the remaining limitations and future extensions. The discussion keeps the claim focused on historical risk-based universe selection rather than final portfolio optimization or guaranteed return improvement.

5.1 Conclusion

This thesis studied asset-universe selection as a decision stage before final portfolio allocation. The central problem was that many allocation methods assume a fixed investable universe, even though the assets admitted into that universe can determine whether the final portfolio is suitable for a conservative, balanced, or aggressive investor. The thesis addressed this gap by building a point-in-time risk-ranking framework that predicts asset-level realized-risk behavior and maps the resulting ranking to investor-profile universes.

For RQ1, the results show that AI/ML can support dynamic asset-universe selection when the task is framed as realized-risk ranking. The promoted PPO model produced positive validation and test ranking quality over a changing Egyptian mixed-asset universe. On the final test split, the ranking remained positive across all test months, and the aggregate test Spearman value indicated meaningful alignment between predicted and realized risk orderings. This supports using the model as an asset-risk ranker for pre-allocation universe selection.

For RQ2, the selected universes aligned with the intended risk-profile interpretation in the historical evaluation. The conservative bucket had materially lower mean realized risk than the full active universe, while the aggressive bucket had higher mean realized risk. The balanced bucket acted as a broader middle universe with overlap into both adjacent profiles, which matches its intended role as a central participation group rather than a strict risk-minimization group. The bucket diagnostics therefore show that predicted ranks can create historically distinct realized-risk groups before allocation.

For RQ3, the thesis used a filter-off full active universe as the empirical baseline. Under the same equal-weight diagnostic rule, the proposed risk-tolerance-based asset universes showed different historical risk characteristics: the conservative universe had lower realized risk than the full active universe, while the aggressive universe had higher realized risk and higher cumulative return in the short test window. This answers RQ3 as a baseline comparison between the full universe and the PPO-filtered risk-tolerance universes. The literature comparison remains important context, but it is not treated as a one-to-one empirical baseline because related methods usually filter by return prediction, price movement, Sharpe-like performance, efficiency, robustness, or final optimizer quality rather than direct realized-risk suitability.

The main contribution is therefore a thesis-safe pre-allocation framework: a leakage-controlled data pipeline, a composite realized-risk target, a masked PPO actor-critic ranker for variable active universes, and an evaluation design that separates selection diagnostics from allocation claims. The contribution is not that the model guarantees better returns or solves final portfolio optimization. It is that the promoted model creates historically distinct realized-risk buckets from predicted ranks and provides evidence that risk-tolerance-oriented universe construction can be studied before final weighting.

Several limitations define the scope of this conclusion. The evaluation is limited to the Egyptian market, so the findings should not be generalized automatically to other markets. Several non-equity exposures are also represented through benchmark or proxy series, including money-market exposure, government bonds, REIT exposure, and gold. The data used in the thesis ends in January 2026. Finally, the main objective is risk-suitability separation, especially reducing realized-risk exposure for the conservative universe, rather than increasing returns for the medium- or high-risk universes. These limitations keep the final claim narrow: the thesis demonstrates historical risk-ranking and selected-universe separation, not guaranteed outperformance.

5.2 Future Work

Future work can extend the thesis in four directions. First, the data pipeline can be connected to live market-data APIs so that daily prices are collected continuously and the model's monthly feature set is rebuilt as new data becomes available. This would allow the promoted risk-ranking framework to be monitored in a live forward-testing setting while preserving the point-in-time protocol used in the thesis.

Second, future experiments can add suitability metrics for higher-risk investor universes. The current thesis ranks assets by realized-risk suitability, which is appropriate for separating conservative, balanced, and aggressive universes by risk exposure. A later extension can test whether the medium- and high-risk universes should include additional return-aware diagnostics after the risk bucket has been formed.

Third, the selected universes can be connected to full downstream allocation experiments. For example, a later study could compare equal weighting with mean-variance optimization or another optimizer after the risk-tolerance bucket has already been formed. This should remain separate from the risk-ranking contribution so that future results do not mix universe selection with final weight optimization.

Fourth, the framework can be evaluated in other markets and asset classes. Applying the same point-in-time risk-ranking protocol to larger markets, such as the American market, or to cryptocurrency universes would test whether the approach remains useful when liquidity, volatility, asset availability, and market structure differ from the Egyptian setting.

Appendix

Appendix A

Lists

This appendix lists abbreviations used in the thesis and then provides the generated lists of figures and tables.

List of Abbreviations

AI	Artificial Intelligence
ML	Machine Learning
RL	Reinforcement Learning
DRL	Deep Reinforcement Learning
PPO	Proximal Policy Optimization
EGX	Egyptian Exchange
EGX30	Egyptian Exchange 30 Index
CPI	Consumer Price Index
USD	United States Dollar
EGP	Egyptian Pound
REIT	Real Estate Investment Trust
EGARCH	Exponential Generalized Autoregressive Conditional Heteroskedasticity
MSE	Mean Squared Error
CNN	Convolutional Neural Network
CSV	Comma-Separated Values
OHLC	Open, High, Low, Close
ATR	Average True Range
RSI	Relative Strength Index
SMA	Simple Moving Average
GAE	Generalized Advantage Estimation
CVaR	Conditional Value-at-Risk
LSTM	Long Short-Term Memory
ANFIS	Adaptive Neuro-Fuzzy Inference System
RQ	Research Question

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Bibliography

[1] H. Markowitz, "Portfolio selection," *The Journal of Finance*, vol. 7, no. 1, pp. 77–91, 1952.

[2] F. Black and R. Litterman, "Global portfolio optimization," *Financial Analysts Journal*, vol. 48, no. 5, pp. 28–43, 1992.

[3] R. T. Rockafellar and S. Uryasev, "Optimization of conditional value-at-risk," *The Journal of Risk*, vol. 2, no. 3, pp. 21–41, 2000.

[4] T. Bodnar, M. Lindholm, E. Thorsén, and J. Tyrcha, "Quantile-based optimal portfolio selection," *Computational Management Science*, vol. 18, pp. 299–324, 2021.

[5] V. DeMiguel, L. Garlappi, and R. Uppal, "Optimal versus naive diversification: How inefficient is the 1/n portfolio strategy?," *The Review of Financial Studies*, vol. 22, no. 5, pp. 1915–1953, 2009.

[6] M. López de Prado, "Building diversified portfolios that outperform out of sample," *The Journal of Portfolio Management*, vol. 42, no. 4, pp. 59–69, 2016.

[7] W. Wang, W. Li, N. Zhang, and K. Liu, "Portfolio formation with preselection using deep learning from long-term financial data," *Expert Systems with Applications*, vol. 143, p. 113042, 2020.

[8] Y. Ma, R. Han, and W. Wang, "Portfolio optimization with return prediction using deep learning and machine learning," *Expert Systems with Applications*, vol. 165, p. 113973, 2021.

[9] T. Kaczmarek and K. Perez, "Building portfolios based on machine learning predictions," *Economic Research - Ekonomska Istraživanja*, vol. 35, no. 1, pp. 19–37, 2022.

[10] A. Chaweewanchon and R. Chaysiri, "Markowitz mean-variance portfolio optimization with predictive stock selection using machine learning," *International Journal of Financial Studies*, vol. 10, no. 3, p. 64, 2022.

[11] E. F. E. Atta Mills and S. K. Anyomi, "A hybrid two-stage robustness approach to portfolio construction under uncertainty," *Journal of King Saud University - Computer and Information Sciences*, vol. 34, no. 9, pp. 7735–7750, 2022.

[12] M. M. Hosseinzadeh, S. Ortobelli Lozza, F. Hosseinzadeh Lotfi, and V. Moriggia, "Portfolio optimization with asset preselection using data envelopment analysis," *Central European Journal of Operations Research*, vol. 31, pp. 287–310, 2023.

[13] F. Abdi, S. Abolmakarem, A. Karbassi Yazdi, Y. Tan, and I. A. Marchioni Choque, "Prospective portfolio optimization with asset preselection using a combination of long and short term memory and sharpe ratio maximization," *IEEE Access*, vol. 12, pp. 144980–144995, 2024.

[14] J.-S. Chou and T.-B.-Q. Pham, "Leveraging ensemble learning-based stock preselection with multiobjective investment optimization for stepwise decision-supported portfolio management," *Journal of Big Data*, vol. 12, p. 196, 2025.

[15] C. Musto, G. Semeraro, P. Lops, M. de Gemmis, and G. Lekkas, "Personalized finance advisory through case-based recommender systems and diversification strategies," *Decision Support Systems*, vol. 77, pp. 100–111, 2015.

[16] A. Asemi, A. Asemi, and A. Ko, "Adaptive neuro-fuzzy inference system for customizing investment type based on the potential investors' demographics and feedback," *Journal of Big Data*, vol. 10, p. 87, 2023.

[17] A. Asemi, A. Asemi, and A. Ko, "A model for investment type recommender system based on the potential investors based on investors and experts feedback using ANFIS and MNN," *Journal of Big Data*, vol. 11, no. 1, p. 128, 2024.

[18] J. Wei and J. Liu, "Personalized fund recommendation with dynamic utility learning," *Financial Innovation*, vol. 11, p. 48, 2025.

[19] H. Alsabah, A. Capponi, O. Ruiz Lacedelli, and M. Stern, "Robo-advising: Learning investors' risk preferences via portfolio choices," *Journal of Financial Econometrics*, vol. 19, no. 2, pp. 369–392, 2021.

[20] C. Yu and Y. Liu, "A personalized mean-CVaR portfolio optimization model for individual investment," *Mathematical Problems in Engineering*, vol. 2021, p. 8863597, 2021.

[21] A. Capponi, S. Ólafsson, and T. Zariphopoulou, "Personalized robo-advising: Enhancing investment through client interaction," *Management Science*, vol. 68, no. 4, pp. 2485–2512, 2022.

[22] C. J. Schneider and Y. Yilmaz, "Stock portfolio selection based on risk appetite: Evidence from ChatGPT," *Finance Research Letters*, vol. 82, p. 107517, 2025.

[23] Y. Deng, F. Bao, Y. Kong, Z. Ren, and Q. Dai, "Deep direct reinforcement learning for financial signal representation and trading," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 28, no. 3, pp. 653–664, 2017.

[24] Z. Jiang, D. Xu, and J. Liang, "A deep reinforcement learning framework for the financial portfolio management problem," *arXiv preprint arXiv:1706.10059*, 2017.

[25] X.-Y. Liu, H. Yang, Q. Chen, R. Zhang, L. Yang, B. Xiao, and C. D. Wang, "FinRL: A deep reinforcement learning library for automated stock trading in quantitative finance," *arXiv preprint arXiv:2011.09607*, 2020.

[26] G. Lucarelli and M. Borrotti, "A deep Q-Learning portfolio management framework for the cryptocurrency market," *Neural Computing and Applications*, vol. 32, pp. 17229–17244, 2020.

[27] F. Soleymani and E. Paquet, "Financial portfolio optimization with online deep reinforcement learning and restricted stacked autoencoder–DeepBreath," *Expert Systems with Applications*, vol. 156, p. 113456, 2020.

[28] M. Pinelis and D. Ruppert, "Machine learning portfolio allocation," *The Journal of Finance and Data Science*, vol. 8, pp. 35–54, 2022.

[29] M. Rezaei and H. Nezamabadi-Pour, "A taxonomy of literature reviews and experimental study of deep reinforcement learning in portfolio management," *Artificial Intelligence Review*, vol. 58, p. 94, 2025.

[30] H. Choudhary, A. Orra, K. Sahoo, and M. Thakur, "Risk-adjusted deep reinforcement learning for portfolio optimization: A multi-reward approach," *International Journal of Computational Intelligence Systems*, vol. 18, p. 126, 2025.

[31] A. Orra, A. Bhambu, H. Choudhary, M. Thakur, and S. Natarajan, "Deep reinforcement learning for investor-specific portfolio optimization: A volatility-guided asset selection approach," *arXiv preprint arXiv:2505.03760*, 2025.

[32] D. B. Nelson, "Conditional heteroskedasticity in asset returns: A new approach," *Econometrica*, vol. 59, no. 2, pp. 347–370, 1991.

[33] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv preprint arXiv:1707.06347*, 2017.